

Jonathan Kim

Summer 2026 Intern - Embedded Software Engineer

Jonathankim829@gmail.com | [Portfolio Page](#) | (909) 657-6129

Education

Master of Science - Embedded Systems

University of California, Irvine

Sept. 2025 - Expected Dec. 2026

Irvine, California

Courses: *Embedded System Modeling and Design, Internet of Things (IoT) Systems and Software*

GPA: (4.0/4.0)

Bachelor of Science - Management Information Systems

Santa Clara University

Sept. 2018 - June 2022

Santa Clara, California

Professional Experience

Embedded Software Engineer Intern

Sept. 2025 - Present

Mirae Opus

Diamond Bar, California

- Developed bare metal C firmware on STM32 microcontroller, including custom I2C and UART drivers and IMU sensor integration.
- Designed and debugged real-time embedded systems using FreeRTOS, including task scheduling, interrupts, and inter-task communication.
- Implemented BLDC motor control for Hall-effect encoders, including a closed-loop PID control system for speed and position control.
- Designed schematics and 4-layer PCB layout, including battery management circuitry, impedance control, and ESD, hot-plug, and over-current protection features.
- Designed a system where a Linux-based controller communicated via UART with an STM32 for real-time control tasks.

Full Stack Software Developer

Jan. 2023 - Oct. 2024

Leidos QTC Health Services

San Dimas, California

- Developed and maintained .NET enterprise applications vital to the business's core revenue stream, contributing to revenue growth from \$500 million to over \$1 billion annually. As a member of the team responsible for core user-facing applications, I ensured seamless functionality during this period of rapid expansion.
- Developed a robust backend using C#, frontend in JavaScript/jQuery, enhancing user experience for scheduling medical appointments.
- Integrated a PDF compression service, significantly improving document processing speeds and resolving critical business bottlenecks.
- Developed 50% of the code for a feature that digitized labor-intensive medical paperwork, previously done manually. This improvement has enhanced the efficiency of care delivery for tens of thousands of veterans.

Projects

STM32 Sentry Turret

April 2025

- Built a sentry turret using a STM32, no HAL library, direct bit register configuration.
- Wrote and implement own drivers for PWM, UART, I2C, and interrupts without any libraries.
- Controlled turret through interrupt-based UART wireless keyboard commands.
- Modified turret gun, designed turret in Fusion 360, 3D-printed and assembled structure.

Technologies & Languages

Technologies: FreeRTOS, UART, I2C, SPI, Linux, Kicad, Fusion 360, STM32, ESP32

Languages: C/C++, Python, C#, HTML/CSS, JavaScript, SQL.